

AfyaPlus Triage: Cost Brief and Go/No-Go

Recommendation: Go, as a pilot, with conditions

Launch triage on the hosted gpt-4o-mini API as a pilot of up to 100,000 requests a month, once the launch checks below pass. Do not self-host an open-source model: at our volume it would cost about 65 times more, and only breaks even at around 8.6 million requests a month.

Cost and spike plan

Monthly	Normal	10x spike
Requests	100,000	1,000,000
Model and hosting fees	\$20	\$230
Staff: 8 hours at \$50 an hour	\$400	\$400
Total	\$420	\$630
Per 1,000 requests, all costs	\$4.20	\$0.63

Model and hosting fees alone are \$0.20 per 1,000 requests. The spike case assumes 5% extra requests from retries and higher hosting overhead. Cloud budget alerts at \$400 a month email finance when spend passes 80% and when it is on track to pass 100%.

Two cost levers

- Answer reuse: an identical message within 10 minutes gets the earlier answer, with no new model call. Cut model calls by 60% on our test set (\$0.20 to \$0.08 per 1,000). Real patient messages repeat less, so expect a smaller saving.
- Grouping: non-urgent summary jobs go to the model 8 at a time; triage is never grouped. Cut calls from 100 to 13 (\$0.11 to \$0.05 per 1,000).

Key risk: a missed emergency

We have not yet tested how often the real model judges urgency correctly, and answer reuse could repeat a wrong answer. Mitigation: before launch the model must flag all 7 emergencies in a 40-message test set reviewed by a clinician. Reused answers expire after 10 minutes and are cleared whenever the model's instructions change.

Conditions

Launch checks: emergency test passed; budget alerts live; a spending limit set on the model provider account, since cloud budgets do not see its fees; fallback message tested.

Targets (proposed, confirm in month one): spend under \$400 a month; p95 response time under 5 seconds, meaning 95 of every 100 requests are answered faster; service available 99.5% of the time.

Fallbacks: if the model provider is down, show a fixed message directing patients to their clinic, or emergency services if symptoms are severe. At the 80% budget alert, pause experiments and summary jobs; triage keeps running.

Stop triggers: any confirmed missed emergency, or spend over budget two months running.

Estimates, not measurements: hosting overhead, staff time, spike retries and self-hosting cost.

The reuse rate comes from test data, not live traffic. Model prices are list prices and should be rechecked before any external quote.